Crypto Funding Carry

Strategy report · Research desk · 12 November 2026

Status: killed, ambiguous test zone · *full-sample t 1.70*

Abstract

The strategy holds Binance USDT perpetuals from Sunday to Sunday: short the names where leveraged longs pay the highest funding, long those where they pay the least, about fifteen names a side. Over the full sample, from 2021 to September 2026, its net Sharpe is 0.75, with a 90 % range from −0.02 to 1.55. On the test, opened on 2025-01-01 data, the net Sharpe is −0.14, with a range from −1.15 to 0.77. **The price P&L, which carried train and validation, flipped sign in the test (the long leg lost 68 % of capital), the full-sample P&L rests on 21 days (87 % of it), and the test break-even cost is 6.9 bp against the 5 to 20 bp assumed.** The strategy is killed: the Analyst killed it before the test on a criterion written in advance, and the test did not lift the kill.

1 Hypothesis and method

Two payers, both the desk's hypotheses. Leveraged longs crowd into the high-funding perps over the weekend, while basis and market-making capital is thin; when that capital comes back on Monday, these names fall against the low-funding ones. And funding itself: longs pay to hold their leverage. The sources show how funding is set from the gap between mark and index (Ackerer 2024, theory) and that carry pays in nine asset classes, crypto not among them (Koijen 2013). No source covers the weekend payer: it was named by the desk after the Red team found that the price P&L arrived in the first two days after the trade.

The universe is the top 30 Binance USDT perpetuals by 30-day quote volume, point in time, listed for at least 90 days, with dead names kept. Each Sunday the book is long the low-funding names and short the high-funding ones, about fifteen a side, held for a week with funding paid or received every eight hours. Costs are set per leg: 20 bp on the short side, 5 bp on the long side, and 12.5 bp on a delisting exit or a liquidation. The train runs from 2021 to 2023, the validation is 2024, and the test runs from 2025-01-01; it was opened on 2026-10-25, after the Analyst had killed the sheet.

2 Performance

The test is the honest estimate: an annual return of −2.4 % at 17.1 % volatility, a net Sharpe of −0.14, and a bootstrap range from −1.15 to 0.77 that covers zero and much of the train result. The full sample looks better only because it holds the train, where the strategy was designed: train net Sharpe 0.89, validation 1.73. The years show the regime: 64 % in 2021, −17 % in 2022, 32 % in 2024, then −3 % in 2025 and −1 % in 2026. The maximum drawdown is −27.9 %, from December 2021 to December 2022, with 673 days under water.

	Sample	5th percentile	Median	95th percentile
Annualised return	14.4 %	−0.4 %	14.3 %	30.0 %
Annualised volatility	19.2 %	18.1 %	19.2 %	20.4 %
Sharpe ratio	0.75	−0.02	0.75	1.55
Maximum drawdown	−27.9 %	−64.9 %	−35.2 %	−21.0 %
Worst month	−20.9 %	−23.3 %	−14.9 %	−9.2 %
Positive months	64 %	49 %	58 %	68 %

Table 1: Net performance, full sample, January 2021 to September 2026, after costs and funding. Percentiles of a stationary block bootstrap of daily returns (blocks of 20 days, 2 000 paths). Returns in fraction of capital, not compounded.

	Sample	5th percentile	Median	95th percentile
Annualised return	−2.4 %	−18.6 %	−3.2 %	13.4 %
Annualised volatility	17.1 %	15.6 %	17.0 %	18.4 %
Sharpe ratio	−0.14	−1.15	−0.18	0.77
Maximum drawdown	−19.9 %	−41.9 %	−22.6 %	−12.4 %
Worst month	−6.9 %	−10.2 %	−7.2 %	−4.7 %
Positive months	52 %	29 %	48 %	62 %

Table 2: Net performance over the test alone, January 2025 to September 2026. Same method as Table 1.

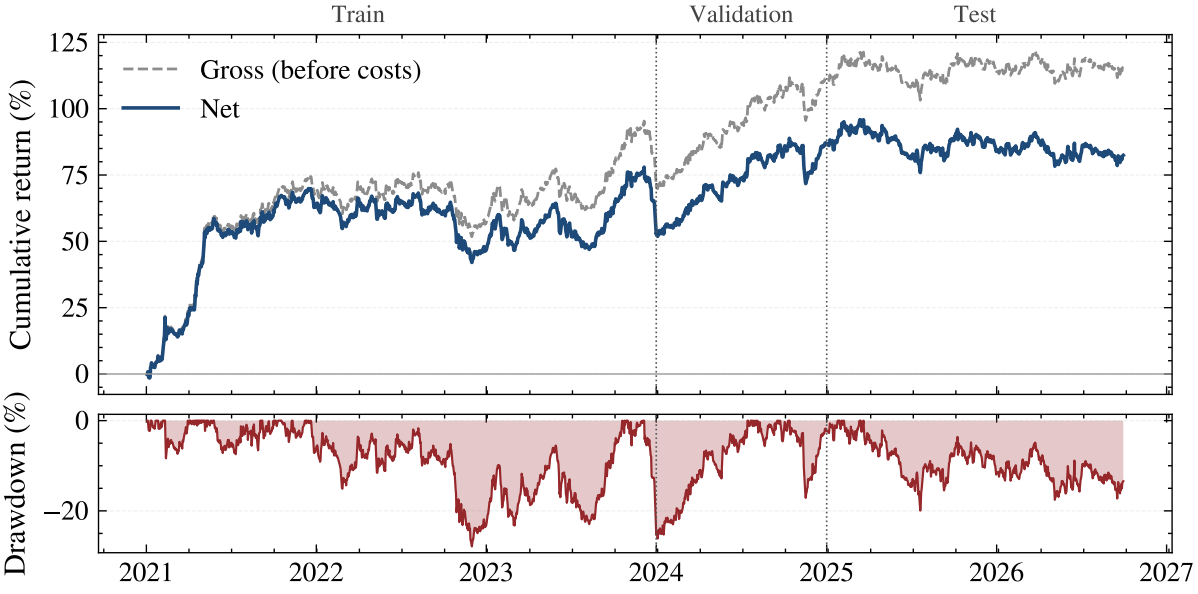


Figure 1: Cumulative net and gross return (top) and drawdown of the net return (bottom), in percent of capital, linear scale. Dotted lines mark the start of the validation and of the test.

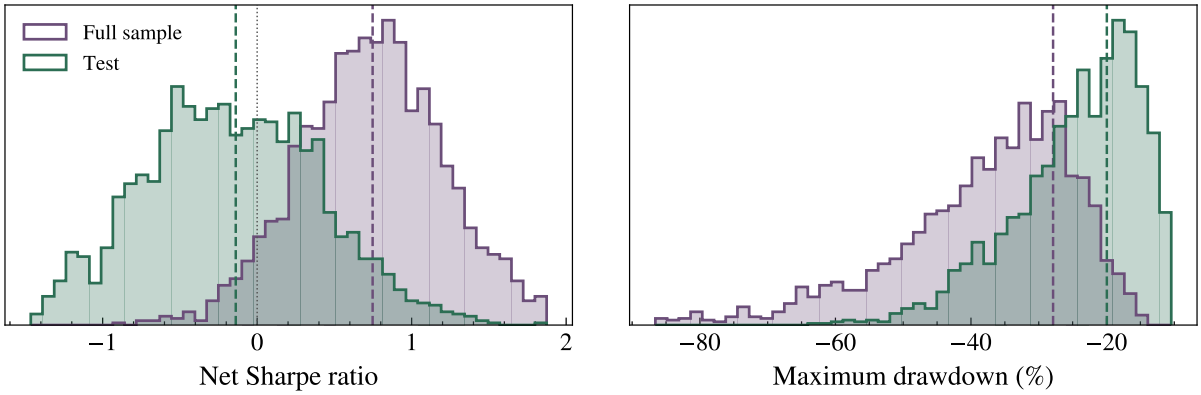


Figure 2: Bootstrap distributions of the net Sharpe ratio (left) and of the maximum drawdown (right). Dashed lines: the values measured on the sample.

3 Market exposure

Against BTC the beta is −0.01 on the full sample and −0.04 on the test, with correlations near zero. The strategy is not a bet on the market. Its alpha over BTC is 14.6 % a year on the full sample with a t of 1.72, and −2.4 % on the test with a t of −0.21: the alpha that looked real before the test is not there after it.

	Full sample	Test
Beta to BTC (daily)	−0.01 (t −0.78)	−0.04 (t −2.37)
Correlation to BTC	−0.02	−0.10
Alpha over BTC, annualised	14.6 % (t 1.72)	−2.4 % (t −0.21)

Table 3: Exposure to the market. Regression of daily net returns on BTC daily returns; t-statistics with Newey-West standard errors (10 lags).

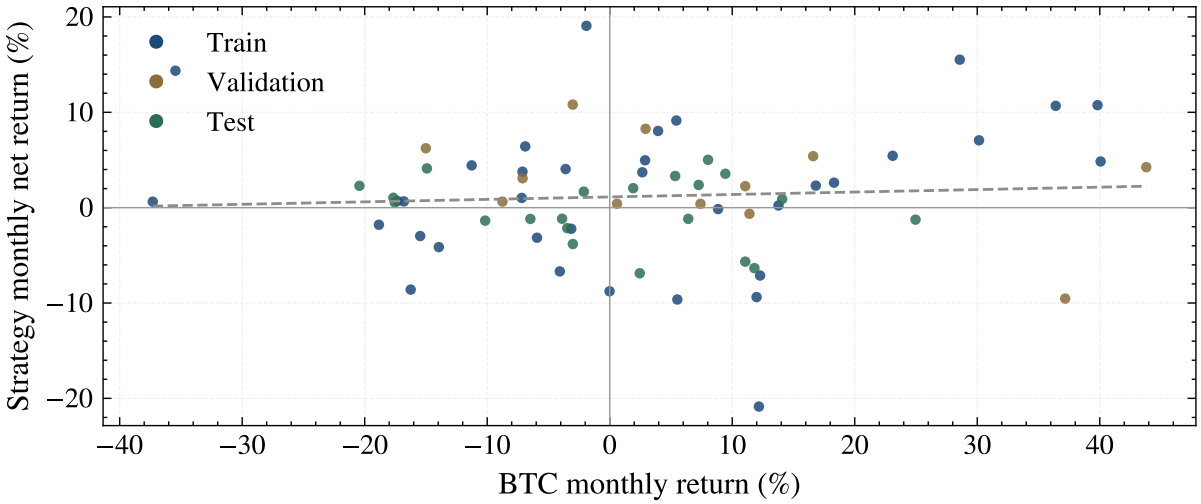


Figure 3: Monthly net return of the strategy against the monthly return of BTC. Dashed line: least-squares fit.

4 Where the profit comes from

In the train, the long leg made 80 % of capital in price and the short leg lost 54 %, with funding adding 47 %. In the validation the long leg still led (46 %) and funding added 7 %. In the test the legs reversed: the long leg lost 68 % and the short leg made 35 %, while funding paid 38 %, more per week than before. Funding, the second payer, paid. The weekend price effect, named as the first payer, did not: it was the source of the earlier P&L, and it flipped.

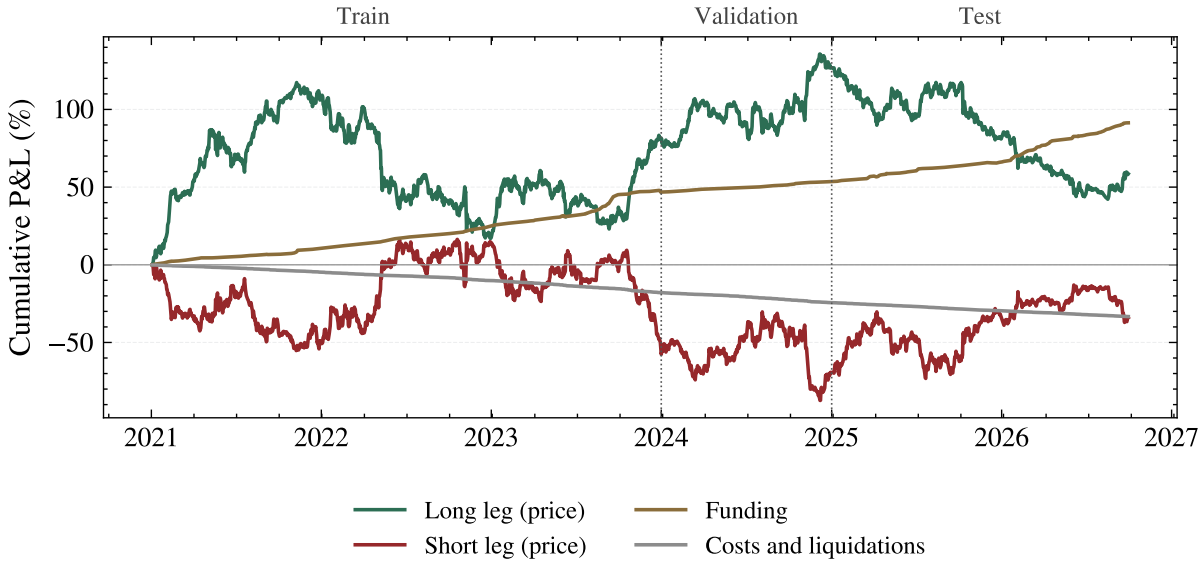


Figure 4: Cumulative P&L by source, in percent of capital: price of the long leg, price of the short leg, funding, and costs.

5 Concentration

On the full sample, the best 1 % of days (21 days) carry 87 % of the net P&L, and without them the Sharpe falls from 0.75 to 0.11; without the best 5 % it is −1.75. BTC itself shows the same nature (Sharpe 0.58, −0.15 without its best 1 % of days), so part of this is crypto. But only 30 % of months stay positive once their two best days are removed. By name, the five best names make 107 % of the train P&L, and without them the train is −5 % of capital. In the test the best name, WLD, makes 206 % of a negative total, and without the best five the test loses 33 %.

	Full sample	Test
Net Sharpe ratio, all days	0.75	−0.14
Share of net P&L from the best 1 % of days	87 % (21 days)	−482 % (7 days)
Net Sharpe ratio without the best 1 % of days	0.11	−0.86
Annualised return without the best 1 % of days	1.9 %	−13.9 %
Net Sharpe ratio without the best 5 % of days	−1.75	−2.78
BTC Sharpe ratio, all days / without its best 1 %	0.58 / −0.15	

Table 4: Concentration of the gains in time. A day removed is set to zero, as if the strategy had been flat that day. The BTC line gives the same cut on the market over the same span.

	Train	Validation	Train and validation	Test
Names traded	137	96	178	182
Share of net P&L, best name	42 %	17 %	33 %	206 %
Share of net P&L, best 5 names	107 %	76 %	78 %	811 %
Share of net P&L, best 10 names	161 %	123 %	124 %	1346 %
Net P&L without the best 5 names, % of capital	−5 %	9 %	24 %	−33 %

Table 5: Concentration of the gains across names: net P&L by name, best names first. A share above 100 % means the other names lost money together.

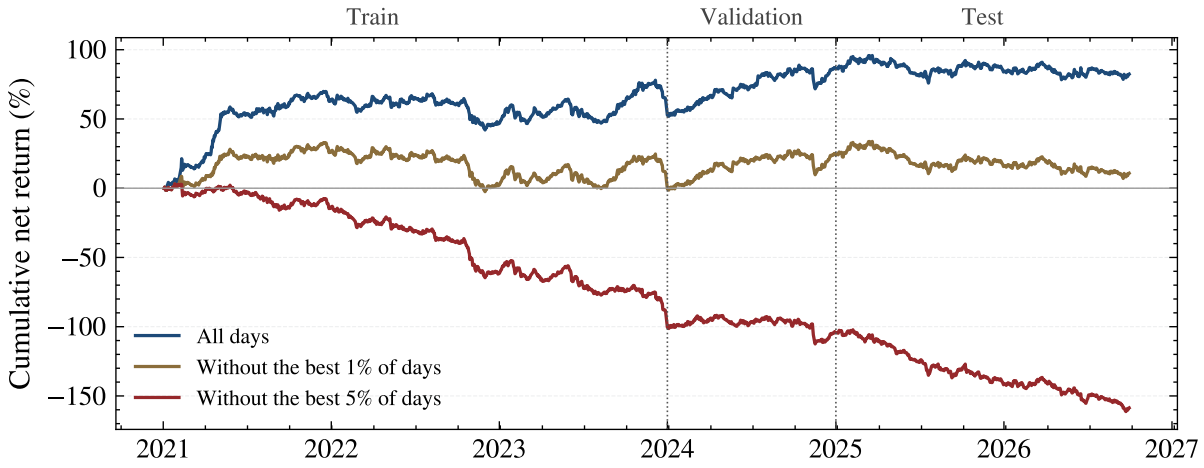


Figure 5: Cumulative net return with all days, and with the best 1 % and 5 % of days set to zero.

6 Implementation

The book holds about fifteen names on each side, with a gross exposure near 1 and an average holding of 60 to 80 days per name. Turnover is about 40 to 48 times a year, for a cost drag of 5 % to 6 % a year. The break-even cost was 55 bp on the train and 80 bp on the validation, but only 6.9 bp on the test, inside the 5 to 20 bp the sheet assumed: the edge left in the test does not cover its own costs.

	Train	Validation	Test
Names held long / short, average	15 / 15	15 / 15	15 / 15
Gross exposure, average	1.08	1.02	0.89
Holding period of a position, days	80	61	61
Turnover, times the capital a year	44.0	47.5	40.4
Cost drag, % of capital a year	−5.7 %	−6.1 %	−5.0 %
Break-even cost, bp one way	55.3	79.7	6.9
Days with a liquidation	22	5	13

Table 6: Implementation. The break-even cost is the one-way cost per unit traded at which the gross P&L of the period would be fully spent.

7 Robustness

Before the test, the result survived higher costs (20 bp on both sides: 0.74 on the train) and every universe and band choice, with train-plus-validation Sharpes from 0.98 to 1.51. Two runs said otherwise. Trading one day late cut the train Sharpe from 0.89 to 0.57, and the same book built on today's top 30 names fell to 0.18: a fast effect and a universe effect. The kill criterion written for the weekend effect, the Monday-Tuesday price P&L without the five best names, fell from 1.31 to 0.48 in Sharpe once those names were removed, a t of 0.98 against a bar of 1.

	Train	Train and validation	Δ train	Δ train and validation
Frozen version, as tested	0.89	1.08		
Cost 5.0 bp		1.34		+0.27
Cost 20.0 bp	0.88	1.00	−0.01	−0.07
Trades 1 day later	0.57		−0.32	

Table 7: Stress tests of the frozen version, run by the Red team before the test: net Sharpe ratio, and its change from the frozen version on the same period.

	Kind	Run on	Net Sharpe	Frozen version
v1	variant	Validation	2.58	1.83
v2	variant	Validation	−0.09	1.83
v3	variant	Validation	2.08	1.83
base-top20	variant	Train and validation	1.38	1.08
base-top50	variant	Train and validation	1.51	1.08
base-age60	variant	Train and validation	1.22	1.08
base-age180	variant	Train and validation	1.05	1.08
base-fw3	variant	Train and validation	1.25	1.08
base-fw30	variant	Train and validation	0.98	1.08
base-band0.0	variant	Train and validation	1.28	1.08
base-band0.2	variant	Train and validation	1.20	1.08
base-todaytop	variant	Train and validation	0.18	1.08
fill00h	variant	Train	1.12	0.89
fill01h	variant	Train	1.15	0.89
fill04h	variant	Train	1.11	0.89
p1-sunday48	variant	Train	1.54	0.89
p1-wed48	variant	Train	0.18	0.89
stress-both20	variant	Train	0.74	0.89
stress-short40	variant	Train	0.66	0.89
stress-short40	variant	Train and validation	0.84	1.08
crit3-04h-all	variant	Train and validation	1.31	1.08
crit3-04h-ex5	variant	Train and validation	0.48	1.08
p1-residual-48h	variant	Train	1.01	0.89

Table 8: Other versions run before the test, each on the period where the desk ran it: variants considered at the validation, controls and diagnostics that check the mechanism. They are not the strategy; the last column gives the frozen version on the same period for comparison.

8 Luck and multiple testing

Six tries were counted for this strategy, with a family-and-asset luck line of 0.41 over two years of train. The train net Sharpe of 0.89 sits 0.48 above it, and the probability that the true train Sharpe exceeds the line is 79 %. That line applies to the train, where the choices were made; the test, which no choice touched, came out at −0.14.

	Value
Tries counted for this strategy	6
Tries in the family, carry on crypto (effective number)	6 (2.1)
Luck line: best Sharpe of 2.1 tries with no skill, 2.0 years	0.41
Train net Sharpe ratio / minus the luck line	0.89 / 0.48
Probability that the true train Sharpe exceeds the luck line	79 %
Whole desk: tries (effective number) / luck line	39 (9.7) / 0.83

Table 9: Multiple testing. Luck line: expected best annual Sharpe ratio of N_{eff} independent tries with no skill (Bailey and López de Prado, 2014); N_{eff} discounts tries by their correlation before costs. The probability is the deflated Sharpe ratio, corrected for the skewness (−0.1) and the kurtosis (7) of daily train returns.

9 Risks and limits

The backtest has three weaknesses the test confirmed. The result rests on a few days and a few names, many of them 2021 alt pumps. The universe matters: today's top 30 would not have made the same money. And the price effect lives in the first one or two days after a Sunday trade, so it depends on the fill time and on weekend liquidity, both of which can change. The short side carries negative skew in squeezes; the test had 13 liquidations and 5 delisting exits.

10 Decision

Killed. The Analyst killed the sheet before the test because its own kill criterion failed (t 0.98 against 1), and a failed criterion kills. The test was then opened by the user and came out in the ambiguous zone with a negative mean (−0.045 % a week) and a full-sample t of 1.70, under 2: it could not lift the kill, and would have allowed paper only at best. The funding leg alone paid in the test; a carry-only version with a stability rule would be a new sheet and a new try, worth revisiting after a year of new data.

A Monthly returns

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2021	5.0	10.7	7.1	19.1	14.4	−3.2	2.6	0.2	3.8	4.8	1.0	−1.8	**63.7**
2022	0.7	−7.1	9.1	0.7	−3.0	0.6	2.3	−4.1	−2.2	−9.6	−8.6	4.0	**−17.2**
2023	10.8	−8.8	5.4	3.7	6.4	−9.4	−6.7	4.4	8.0	15.5	−0.2	−20.9	**8.5**
2024	0.4	4.2	5.4	6.2	−0.6	3.1	8.3	0.6	0.4	2.3	−9.5	10.8	**31.6**
2025	3.6	1.0	1.7	0.9	−5.7	−6.9	5.0	−1.2	3.3	−1.2	0.6	−3.8	**−2.6**
2026	−1.4	4.1	2.0	−6.3	−2.2	2.3	2.4	−1.3	−1.2				**−1.5**

Table 10: Monthly net return, % of capital.

Runs: 20261025-154720-train-base, 20261025-154721-validation-base, 20261025-153304-train+validation-base, 20261025-154903-test-base. Strategy file: crypto_funding_carry.py. Frozen version: base.